Spring 2021, MATH 8120 Real Analysis II, Final Exam

PRINT NAME:______________________________ **ID#:** ______________________

- Show your work. Answers with no or incorrect steps will not receive partial credit.
- The exam is self-explanatory. The instructor will not interpret the questions.

1. (5 points) Let Ω be an open bounded subset of $\mathbb{R}^n$ and $p \in (1, \infty)$. Let $\{f_k\}$ be a sequence of bounded functions in $L^p(\Omega)$ and $f_k \to f$ pointwisely for some $f \in L^p(\Omega)$. Show that $f_k \rightharpoonup f$ in $L^p(\Omega)$.

   Proof. Note that $L^p(\Omega)^* = L^q(\Omega)$. For any $g \in L^q(\Omega)$, we have $|f_k g| \to |fg|$ a.e. as $k \to \infty$. Let $M > 0$ be such that $|f_k(x)| \le M$ for all x and k (since $\{f_k\}$ are bounded). Then $|f_k g| \le M|g|$ which is integrable since $g \in L^q(\Omega) \subset L^1(\Omega)$. Hence, by dominated convergence theorem, we know $\int f_k g \to \int fg$. As g is arbitrary, we know $f_k \rightharpoonup f$ in $L^p(\Omega)$. $\square$

2. (5 points) Let X be a Banach space and let $\{f_k\}$ be a sequence in X^* that is weakly* convergent in X^*.
   (a) Show that $\{f_k\}$ is bounded.
   (b) Show that the weak* limit f of $\{f_k\}$ satisfies $\|f\| \le \liminf_{k\to\infty} \|f_k\|$.

   Proof. (a) Since for any $x \in X$, $\{f_k(x)\}$ is a convergent sequence in $\mathbb{R}$, we know $\sup_{k\ge1} |f_k(x)| < \infty$. Then, by Uniform Boundedness Theorem, we know there exists $M > 0$ such that $\|f_k\| \le M$ for all $k \in \mathbb{N}$.
   (b) For any $x \in X$, we have $|f_k(x)| \le \|f_k\|\|x\|$. Moreover
   $$|f(x)| = \lim_{k\to\infty} |f_k(x)| \le \liminf_{k\to\infty} \|f_k\|\|x\| \le M\|x\|.$$
   Hence $\|f\| \le \liminf_{k\to\infty} \|f_k\|$.

   $\square$

3. (5 points) Let $(X, \|\cdot\|)$ be a reflexive Banach space and let Z be a nonempty closed convex subset of X.
   (a) Show that, given any $x \in X$, there exists $y \in Z$ such that $\|x - y\| = \inf_{z\in Z} \|x - z\|$.
   (b) Show that y is unique if $(X, \|\cdot\|)$ is strictly convex.

   Proof. (a) let $y_k \in Z$ be a minimizing sequence, i.e., $\|x - y_k\| \to \inf_{z\in Z} \|x - z\|$. Then $\{y_k\}$ is bounded. By Eberlein-Smulian theorem, there exists a subsequence $\{y_{k_j}\}$ such that $y_{k_j} \rightharpoonup y$ for some $y \in X$. Let $\{z_j\}$ be the convex combinations of $\{y_{k_j}\}$ such that $z_j \to y$. Since $z_j \in Z$ and Z is closed, we know $y \in Z$.
   (b) If not, then there exist distinct $y, y' \in X$ such that $\|x-y\| = \|x-y'\| = \inf_{z\in Z} \|z-x\|$. Then for any $\lambda \in (0, 1)$, we know $\lambda y + (1 - \lambda)y' \in Z$ but
   $$\|x - (\lambda y + (1 - \lambda)y')\| < \lambda\|y\| + (1 - \lambda)\|y'\| = \inf_{z\in Z} \|x - z\|$$
   which is a contradiction.

   $\square$

4. (5 points) Let $(X, \langle\cdot, \cdot\rangle)$ be a Hilbert space and let $T \in L(X)$. Show that, if there exists a constant $\alpha > 0$ such that $\langle Tx, x\rangle \ge \alpha\|x\|^2$ for any $x \in X$, then T is bijective.

   Proof. If $Tx = 0$, then $0 = \langle Tx, x\rangle \ge \alpha\|x\|^2 = 0$, which implies $x = 0$. Since $\alpha\|x\|^2 \le \langle Tx, x\rangle \le \|Tx\|\|x\|$, we know $\|x\| \le \frac{1}{\alpha}\|Tx\|$. Therefore $T(X)$ is a closed linear subspace of X. If there exists $X \setminus T(X)$ is nonempty, then there exists a nonzero $y \in T(X)^\perp$. This implies that $0 = \langle Ty, y\rangle \ge \alpha\|y\|^2$, which is a contradiction. $\square$

5. (5 points) Let X be a normed linear space and let $g : X \to \mathbb{R}$ be a convex and continuous function. Show that there exists $f \in X^*$ and $c \in \mathbb{R}$ such that $g(x) > f(x) + c$ for all $x \in X$.

Proof. Since g is convex and continuous at 0, we know $\partial g(0)$ is nonempty. Let $f \in \partial g(0)$ and $c < g(0)$, then there is

$$g(x) \geq f(x-0) + f(0) > f(x) + c$$

for all $x \in X$ which completes the proof. □

6. (5 points) Let X be a Banach space and $x \in X$. Let $E_x := \{f \in X^* : \|f\| \leq \|x\|,\ f(x) = \|x\|^2\}$. Show that E_x is nonempty, closed, and convex

   Proof. By Hahn-Banach theorem, there exists $f_0 \in X^*$ such that $\|f_0\| = 1$ and $f_0(x) = \|x\|$. Then $f := \|x\| f_0$ is in E_x, which means E_x is nonempty.

   Suppose $f_k \to f$ where $\{f_k\} \subset E_k$, then $\{f_k\}$ is Cauchy in X^* (which is a Banach space). So $f_k \to f \in X^*$ and $\|x\| \geq \|f_k\| \to \|f\|$ which implies $\|f\| \leq \|x\|$ and $\|x\|^2 = f_k(x) \to f(x)$ which means $f(x) = \|x\|^2$. Hence $f \in E_x$, which means that E_x is closed.

   For any $f, f' \in E_x$ and $\lambda \in [0,1]$, there are

$$\|\lambda f + (1-\lambda) f'\| \leq \lambda \|f\| + (1-\lambda)\|f'\| \leq \|x\|,$$
$$(\lambda f + (1-\lambda) f')(x) = \lambda f(x) + (1-\lambda) f'(x) = \|x\|^2.$$

   Hence E_x is convex. □

7. (5 points bonus) Mark each of the following assertions True (T) or False (F). Let X be a Banach space and $T \in L(X)$.

   (a) ☐ If $\{x_k\}$ is bounded in X, then it has a weakly convergent subsequence.

   (b) ☐ If $\{f_k\}$ is bounded in X^*, then it has a weakly* convergent subsequence.

   (c) ☐ The resolvent set $\rho(T)$ of T can be bounded in $\mathbb{C}$.

   (d) ☐ If $T^{-1} \in L(X)$, then there exists $\delta > 0$ such that $S^{-1} \in L(X)$ for all S satisfying $\|T - S\| \leq \delta$.

   (e) ☐ If X^* is reflexive, then so is X.

   (a) False. Need X be reflexive.
   (b) True. Alaoglu theorem.
   (c) False. $\bar{B}(0; r_\sigma(T))^c \subset \rho(T)$ so $\rho(T)$ must be unbounded.
   (d) True. For $\delta < 1/\|T\|$.
   (e) True. X is reflexive if and only if X^* is reflexive.